Innovation Centre for Education

# Yenepoya Institute of Arts Science Commerce and Management

# Final Project Report

# on

# Develop a virtual museum to explore artifacts in 3D

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# Background

**Museums play a crucial role in preserving cultural, historical, and scientific artifacts, but access to physical museums is often limited due to geographical distance, time constraints, and other practical challenges. With the rapid advancement of digital technologies, especially in 3D graphics and interactive systems, it has become possible to recreate real-world environments in virtual spaces.**

**The development of game engines like Unity has made it easier to design immersive and interactive applications that simulate real-life experiences. These technologies enable the creation of virtual environments where users can explore, interact, and learn in an engaging manner.**

**The concept of a Virtual Museum emerges from the need to make museum experiences more accessible and interactive. Instead of being limited to physical visits, users can explore exhibits from anywhere using a digital platform. This approach not only increases accessibility but also enhances learning by providing additional multimedia content such as 3D models, audio explanations, and interactive elements.**

**Furthermore, virtual museums are increasingly being adopted in education and research, as they allow students and learners to experience artifacts in a more dynamic and engaging way. This project builds upon these advancements to create a Virtual Museum that combines technology, education, and user experience into a single platform.**

# Aim

**The aim of this project is to develop an interactive Virtual Museum that allows users to explore artifacts in a three-dimensional environment. The system is designed to simulate a real museum experience using Unity, enabling users to navigate through galleries, interact with exhibits, and view detailed information about artifacts. The project focuses on improving accessibility to cultural and educational content by providing a digital platform that can be accessed anytime and from anywhere.**

# Technologies Used

**The development of the Virtual Museum involves various technologies that work together to create a smooth and interactive experience.**

**Core technologies include:**

- **Unity Engine for 3D development and rendering**
- **C# programming language for scripting and logic implementation**
- **Blender or similar tools for 3D modeling (optional)**

**These technologies ensure efficient performance, realistic visuals, and easy scalability of the system.**

# Hardware Architecture

**The hardware architecture defines the physical components required to run the Virtual Museum system effectively. The system is designed to operate on standard computing devices without requiring specialized hardware.**

**Minimum requirements include:**

- **A computer or laptop with a modern processor (Intel i3 or above)**
- **Minimum 4GB RAM (8GB recommended for better performance)**
- **Graphics support (integrated or dedicated GPU)**
- **Input devices such as keyboard and mouse**

**Optional hardware such as VR headsets can be integrated to enhance user experience and provide immersive interaction.**

# Software Architecture

**The software architecture of the Virtual Museum follows a modular design approach, ensuring that different components of the system are independent and maintainable. The application is primarily built using Unity, which handles rendering, physics, and interaction.**

**The architecture consists of the following layers:**

- **Presentation Layer:**
  **This layer includes the user interface and 3D environment where users interact with the system.**
- **Application Layer:**
  **This layer handles the core logic, including navigation, interaction, and data processing.**
- **Data Layer:**
  **This layer manages storage and retrieval of artifact data using local files or databases.**

**The modular architecture allows easy updates, scalability, and integration of new features without affecting the overall system.**

# System Requirements

**The system requirements define the necessary functionalities, user expectations, and operating conditions required for the successful development and execution of the Virtual Museum. These requirements ensure that the system performs efficiently and meets user needs.**

# Functional Requirements

**Functional requirements describe what the system should do. In the Virtual Museum, the system must allow users to navigate the 3D environment and interact with artifacts seamlessly.**

**The system should:**

- **Allow users to move freely within the virtual museum**
- **Enable selection of artifacts through interaction (click/raycast)**
- **Display detailed information about selected artifacts**
- **Load and render 3D models efficiently**

- **Provide a user-friendly interface for navigation and controls**

## User Requirements

**User requirements focus on the expectations and needs of the end users. The system should be simple, intuitive, and accessible to users with basic computer knowledge.**

**Users expect:**

- **Easy navigation using keyboard and mouse**
- **Clear and readable artifact information**
- **Smooth and lag-free experience**
- **Visually appealing environment**
- **Minimal learning curve for interaction**

## Environmental Requirements

**Environmental requirements define the platform and conditions under which the system operates.**

**Requirements include:**

- **Operating system: Windows (primary)**
- **Development platform: Unity**
- **Hardware: Minimum 4GB RAM and basic graphics support**
- **Input devices: Keyboard and mouse**
- **Optional: VR support for enhanced experience**

## Design and Architecture

**The system follows a modular architecture where each component is designed to perform a specific function. The architecture separates the application into layers such as presentation, logic, and data handling. The navigation system handles user movement, the interaction system detects object selection, and the UI system displays information.**

**This structured design ensures that the system is scalable, maintainable, and easy to update with new features.**

## Implementation

**The implementation of the Virtual Museum is carried out using Unity and C# scripting. The 3D environment is created using models and textures, while scripts are used to control movement, interaction, and UI behavior.**

**The development process includes:**

- **Creating the 3D museum environment**
- **Importing and placing artifact models**
- **Writing scripts for navigation and interaction**
- **Designing UI panels for information display**

## Testing

**Testing is performed to ensure that the system works correctly and efficiently. Different types of testing are carried out to identify and fix errors.**

**Testing includes:**

- **Functional testing (checking features)**
- **Performance testing (checking lag and FPS)**
- **Usability testing (ease of use)**
- **Bug fixing and validation**

## Graphical User Interface (GUI) Layout

**The GUI is designed to provide a simple and intuitive user experience. It includes navigation controls, menus, and information panels that display artifact details.**

**Key elements:**

- **Main menu (start/exit options)**
- **On-screen instructions**
- **Artifact information panel**
- **Minimal and clean design**

**The layout ensures that users can easily interact with the system without confusion.**

## Customer Testing

**Customer testing involves evaluating the system with real users to gather feedback. Users are allowed to explore the virtual museum and provide suggestions based on their experience.**

**Focus areas:**

- **Ease of navigation**
- **Clarity of information**
- **Overall user experience**
- **Performance and responsiveness**

**This feedback helps in improving the system before final deployment.**

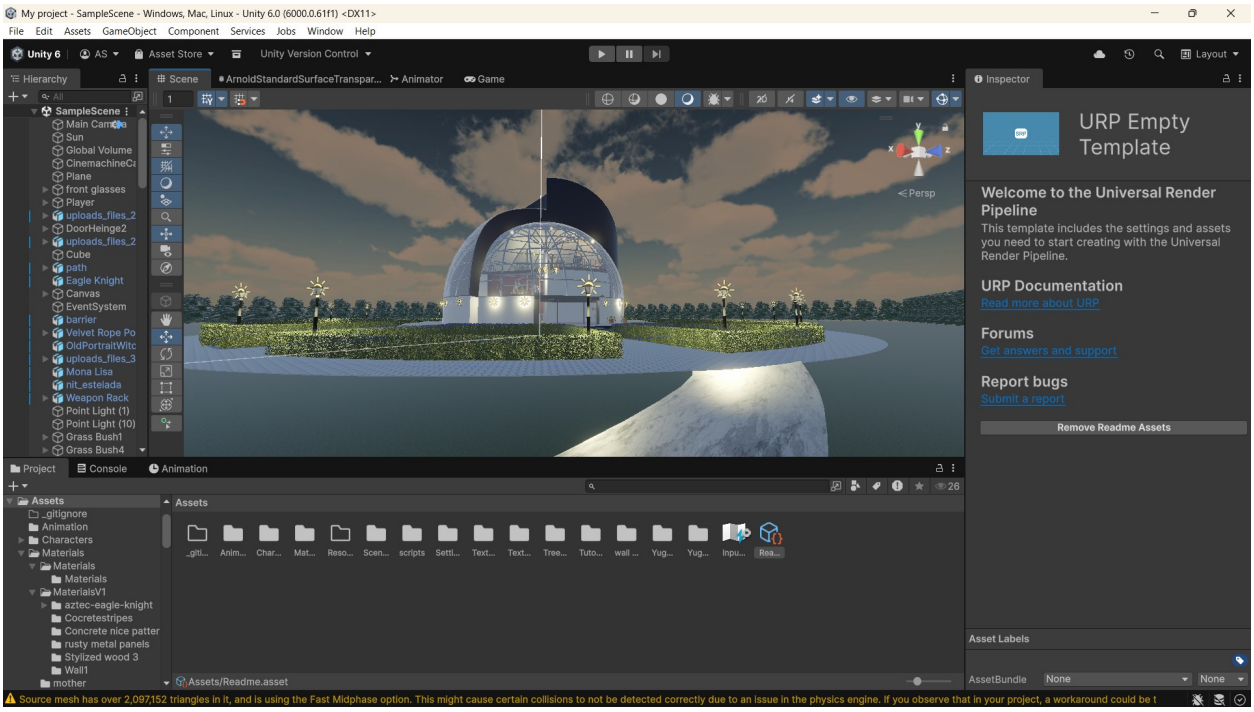
## Evaluation

**The system is evaluated based on its performance, usability, and effectiveness in achieving its objective. The Virtual Museum successfully provides an interactive platform for exploring artifacts and enhances accessibility to museum content.**

**Evaluation criteria:**

- **User satisfaction**
- **System performance**
- **Ease of use**
- **Quality of interaction**

**Overall, the system meets its goals and demonstrates the effective use of 3D technology for educational purposes.**




# 1. Top Bar (Toolbar)

- **Shows your project name and platform: Windows, Mac, Linux**
- **Play ▶, Pause , Step buttons → used to run your scene**
- **You are currently in Edit mode (not playing)**

# 2. Hierarchy Panel (Left Side)

**This shows all objects in your scene (SampleScene).**

Important objects visible:

- **Main Camera → Player's view**
- **Sun → Directional light (daylight)**

- **Global Volume → Post-processing (lighting effects)**
- **Cinemachine → Camera system**
- **Plane → Ground**
- **Player → Your character/controller**
- **front glasses → Glass object (maybe your dome)**
- **uploads_files_2 / models → Imported assets**
- **Point Light → Lights for lamps**
- **Canvas → UI system**

**This is basically your entire museum structure list**

## 3. Scene View (Center)

**This is where your 3D world is visible.**

What we see in your scene:

- **A modern glass dome building (your museum)**
- **Pathway around it**
- **Street lamps (with lighting)**
- **Bushes/greenery**
- **Skybox (cloudy sky)**
- **Water or edge area (looks like island or platform)**

**This is your Virtual Museum environment**

## 4. Game View Tab

- **Next to Scene tab**
- **Shows what player sees during gameplay**
- **Currently not active (you are in Scene view)**

## 5. Project Panel (Bottom)

**This shows all your project files.**

Folders visible:

- **Animation**

- **Characters**
- **Materials**
- **Scenes**
- **Scripts**
- **Textures**

**This is where all your assets are stored**

## 6. Inspector Panel (Right Side)

**Currently showing:**

- **URP Empty Template info**

**Normally this shows:**

- **Object properties (position, material, collider, etc.)**
- **But right now, no object is selected → so it shows template info**

## 7. Lighting & Environment

- **Sky with clouds → Skybox**
- **Lamps → Point lights**
- **Bright reflections → URP lighting**

**You are using URP (Universal Render Pipeline) for better graphics**

## Conclusion

**The Virtual Museum project successfully demonstrates how modern 3D technologies can be used to create an interactive and immersive platform for exploring artifacts. By using Unity, the system provides a realistic environment where users can navigate through virtual galleries, interact with exhibits, and access detailed information in an engaging manner.**

**The project achieves its primary objective of improving accessibility to museum content by allowing users to experience exhibits digitally, regardless of their physical location. It also enhances learning by combining visual elements with interactive features, making the experience more informative and engaging compared to traditional methods.**

**Throughout the development process, key aspects such as performance, usability, and system design were carefully considered. The modular architecture ensures that the system is scalable and can be extended with additional features in the future. Testing and evaluation confirm that the system provides a smooth and user-friendly experience.**

**In conclusion, the Virtual Museum serves as an effective application of technology in the field of education and cultural preservation. It highlights the potential of virtual environments in transforming how users interact with and learn from digital content.**

## Further Development / Reseach

**The Virtual Museum project provides a strong foundation for creating an interactive digital platform; however, there are several opportunities for further development and research to enhance its capabilities and user experience. With advancements in technology, the system can be extended to provide more immersive, scalable, and intelligent features.**

**One major area of improvement is the integration of Virtual Reality (VR) and Augmented Reality (AR) technologies. By using VR headsets, users can experience a fully immersive environment, making the virtual museum feel closer to a real-world visit. Similarly, AR can allow users to project artifacts into their physical surroundings, improving accessibility and engagement.**

**Another important enhancement is the inclusion of multiplayer or collaborative features, where multiple users can explore the museum simultaneously. This would enable guided tours, virtual classrooms, and interactive learning sessions, making the platform more useful for educational purposes.**

**The system can also be improved by adding advanced search and recommendation systems. By incorporating intelligent algorithms, the application can suggest artifacts based on user interests, browsing behavior, or historical relevance. This would make the experience more personalized and efficient.**

**Further development can include integration with cloud-based databases to allow dynamic updates of artifacts without modifying the application. This would enable museum administrators to add, update, or remove exhibits in real time.**

**Additionally, the graphical quality and realism of the system can be enhanced by using advanced rendering techniques, better lighting models, and high-quality textures in Unity. Optimization techniques can also be applied to ensure smooth performance across different devices.**

**From a research perspective, this project can be extended to study user engagement, learning effectiveness, and accessibility in virtual environments. It can also explore how digital museums impact education compared to traditional learning methods.**

**In conclusion, the Virtual Museum has significant potential for growth and innovation. By incorporating emerging technologies and expanding its features, it can evolve into a powerful platform for education, cultural preservation, and interactive learning.**

## References

**The following resources were used during the development of the Virtual Museum project. These include official documentation, tools, and platforms that supported the design, implementation, and research.**


- **Unity Documentation – For understanding 3D development, scripting, and rendering**
- **Blender Documentation – For creating and editing 3D models (if used)**
- **Sketchfab – For downloading 3D assets and models**
- **CGTrader – For obtaining additional 3D models**
- **Unity Asset Store – For textures, materials, and assets**
- **General online tutorials and educational resources related to Unity and 3D application development**

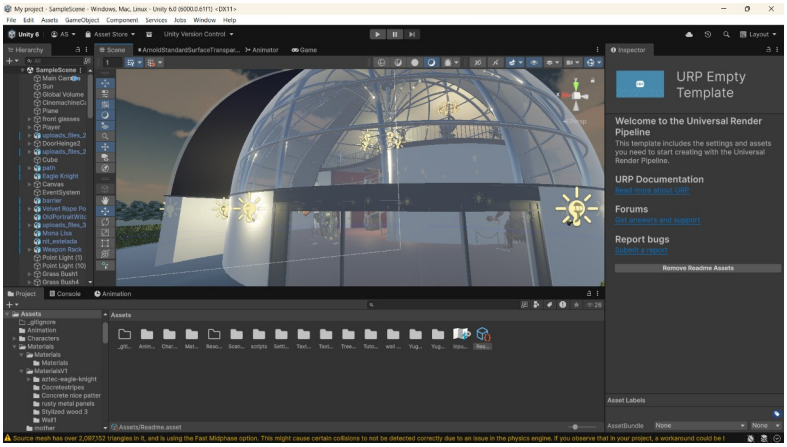
# Appendix

**The appendix contains additional supporting information and materials related to the project. These details provide further clarity on the implementation and development process.**

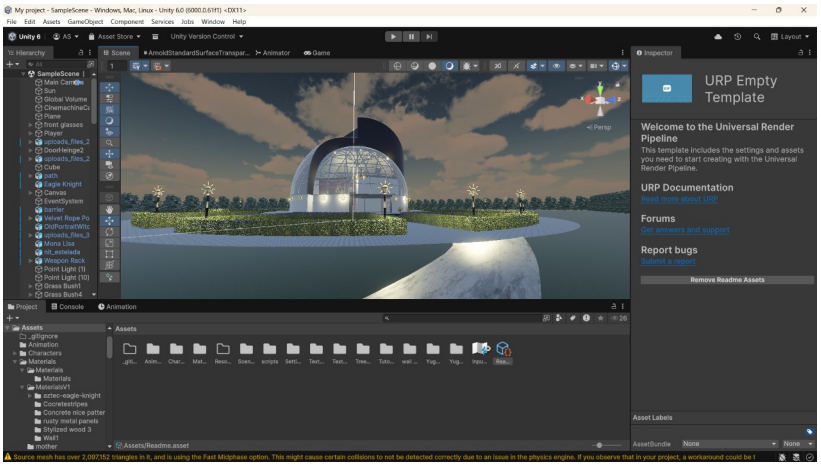
## A. Screenshots of the Application

**This section includes screenshots of different parts of the Virtual Museum such as:**

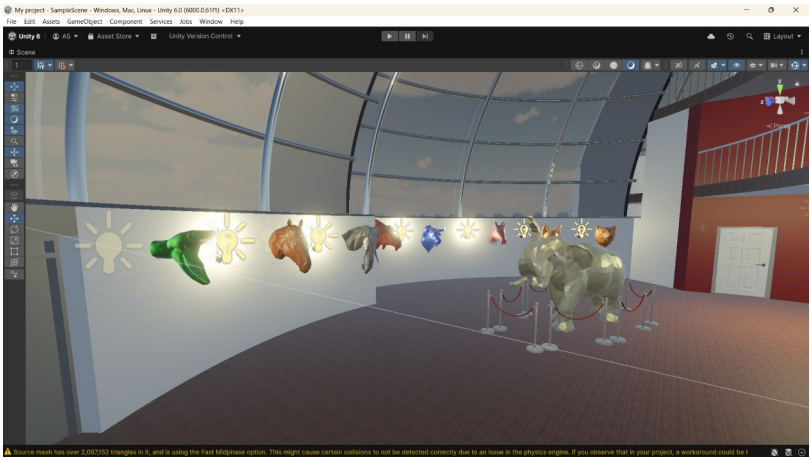
- **Main menu**




- **3D museum environment**



- **Artifact interaction view**




- **Information display panel**

## B. Sample Code Snippets

**Important code snippets used in the project are included here for reference.**

**Examples:**

- **Player movement script**
- **Interaction (raycasting) script**
- **UI display script**

## C. Asset Details

**This section lists the external assets used in the project.**

**Includes:**

- **3D models (source and format)**
- **Textures and materials**
- **Audio or media files (if used)**

## D. Testing Results

**This section provides results from system testing.**

**Includes:**

- **Performance observations**
- **Bug reports and fixes**
- **User feedback summary**

## E. User Guide (Optional)

**This section explains how to use the application.**

**Steps:**

- **Launch the application**
- **Navigate using keyboard and mouse**
- **Select artifacts to view details**
- **Exit the application**